

UTCI Validation + First Graphs

1. Validation of Dev's Real UTCI Output

Ran full sanity checks on `notebooks/data/processed/utci_hourly.csv` (117,936 rows, 2021–2026, Ahmedabad).

Check	Result	Verdict
Temperature (tdb) range	19.53°C to 45.22°C	✓ Shensible for Ahmedabad
Impossible temp values (<−10°C or >55°C)	0	✓ Pass
RH range	10.99% to 99.91%	✓ Within valid 0–100%
Impossible RH values	0	✓ Pass
Wind speed range	0.013 to 13.05 m/s	✓ No negatives
MRT range	6.93°C to 68.97°C	✓ Pass — see note below
Missing values (any column)	0	✓ Pass
Duplicate (time, lat, lon) rows	0	✓ Pass
Timestamp coverage	2021–2026, 2,184 hourly steps/year, consistent	✓ Pass
UTCI range	13.95°C to 51.06°C	✓ Physically sensible, correctly reaches "extreme heat stress" (>46°C) category
Missing/NaN UTCI	0	✓ Pass

Note on MRT minimum (6.93°C): occurs at 1:00 AM on April 1, 2021, with air temperature 22.1°C at the same hour. A 15°C gap between MRT and air temperature at night is physically expected — clear-sky nighttime radiative cooling causes surfaces to lose heat to the sky faster than the air cools, pushing MRT below air temperature. Not a bug.

Conclusion: Dev's pipeline passes every validation check. No data quality issues found.

2. First UTCI Graphs

Graph 1: Hourly UTCI, May 2024 (representative single-month window)

`utci_may2024_timeline.png` — shows the full diurnal cycle, with daily peaks increasingly crossing the 46°C "extreme heat stress" threshold as the month progresses, and nighttime minimums also rising later in the month (from ~20–25°C early May to ~28–30°C by late May). This is the clearest single graph to use for the PPT's "Real Proof of Work" slide — it visually tells the whole NCTL story (rising nighttime floor) even before NCTL is formally calculated.

Graph 2: Daily maximum UTCI, all 6 years (2021–2026)

`utci_6year_overview.png` — every year shows a similar pattern: UTCI climbs above the 46°C extreme-heat-stress threshold during April–June, consistently across all 6 years. This is good supporting evidence for the "pattern holds across multiple years" validation story.

Important side-finding: this graph visually confirms why Dev's "per-year CTBI reset" fix (discussed earlier) was necessary — the straight diagonal lines connecting the end of one year to the start of the next are exactly the artifact that would have corrupted CTBI if computed naively across the merged timeline. Worth keeping this graph in the Decision Log as visual justification for that fix, not just in the PPT.

Handoff

- Both graphs ready for the PPT's "Real Proof of Work" slide
- Validation results go in the Decision Log as evidence the pipeline is clean and checked
- No issues found means Dev can proceed to Day 3 (NCTL/CTBI) on this data with confidence